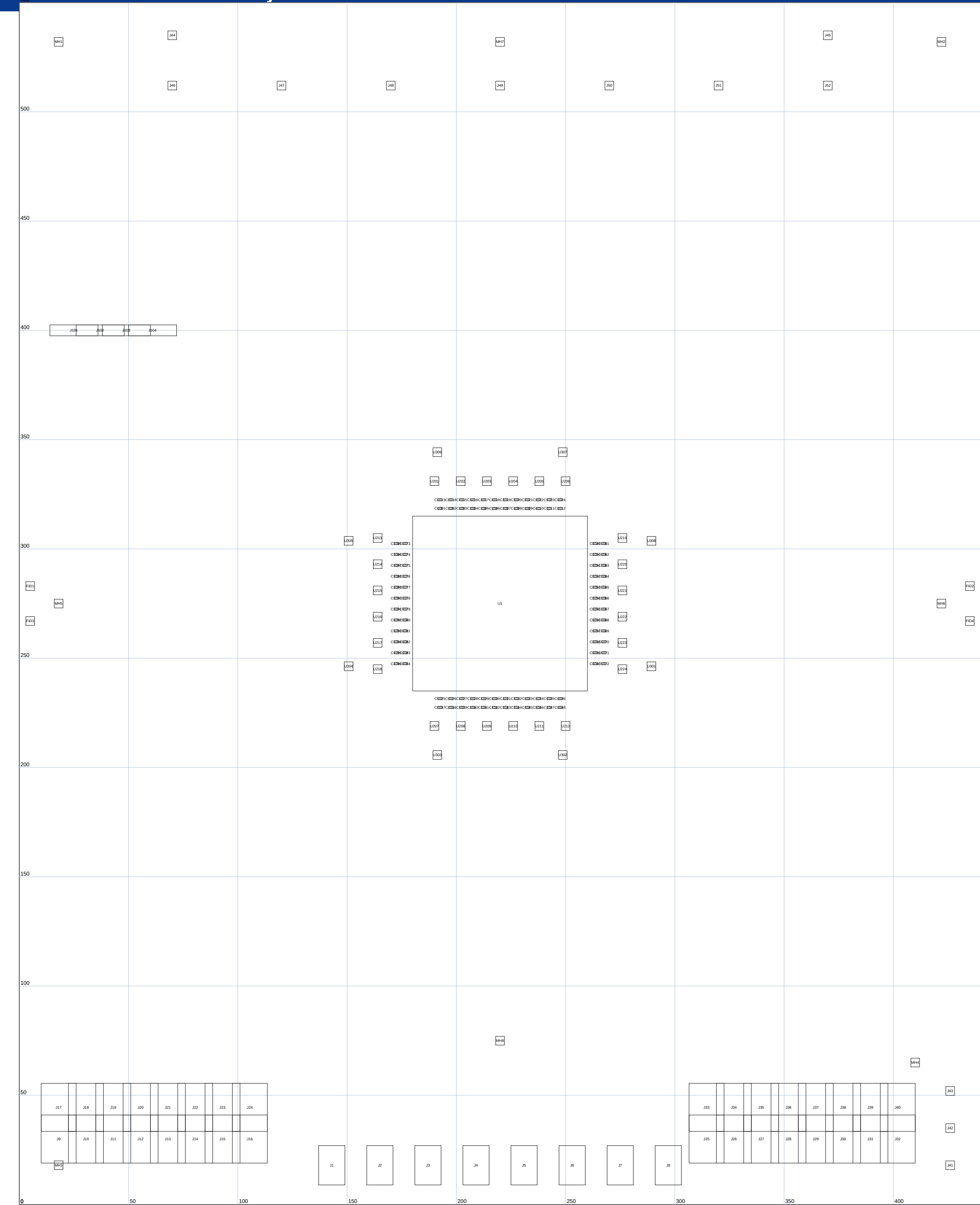




## Layer characteristics

|                  |                                         |
|------------------|-----------------------------------------|
| Layer name       | F.Fab                                   |
| Stack-up index   | 25 of 27                                |
| Role             | Fab assembly                            |
| Net / signal     | —                                       |
| Copper weight    | —                                       |
| Dielectric above | Megtron-7 · Dk 3.40 · Df 0.002 @ 10 GHz |
| Reference layer  | GND                                     |
| Impedance target | —                                       |
| Antipad rule     | 0.20 mm clearance · BGA escape Type-A   |
| Microvia type    | IPC-4761 Type VII · filled-and-capped   |
| Back-drill       | ≤ 6 mil stub · PCIe Gen 6 / HBM4        |
| Trace width      | 4.0 mil / 4.0 mil clearance             |
| Test coupon      | IPC-2141 Appendix B                     |
| AOI inspection   | 100 % bare-board ET + AOI               |
| Fab note ref     | See Fab_Drawing notes 1–18              |

## Compliance summary

- IPC-6012 Class 3/A acceptance, IPC-A-600 Class 3 visual
- IPC-2581 Rev C unified data package included
- IPC-7351B 2D-data-matrix date code on B.Silk
- IPC/JEDEC J-STD-033 MSL-3 packaging
- RoHS / REACH compliant - Pb-free SAC305 reflow
- Conformal coating: none (sealed liquid-cooled chassis)

| LightRail AI — Confidential / Investor data-room |                                  |
|--------------------------------------------------|----------------------------------|
| Programme                                        | LR-S1A 1U CPO Switch Motherboard |
| Drawing No.                                      | LR-S1A-LAY-025                   |
| Layer                                            | F.Fab                            |
| Role                                             | Fab assembly                     |
| Designer                                         | LightRail AI — Design Automation |
| Date / Rev                                       | 2026-04-19 · Rev 1.0             |